

Assignment 1: Rice Classification

Key information

Deadlines

Submission: 11:59 pm, 18 September 2026 (Friday week 7)

Late submissions policy

Late submissions are allowed up to 3 days late. A penalty of 5% per day late will apply. Assignments more than 3 days late will not be accepted (i.e. will get 0 marks). The day cut-off time is 11:59 pm.

Marking

This assignment is worth 15 marks = 15% of your final mark.

Please see the marking rubric on Canvas. Your marks contain three parts, including:

1. Your code: Most of the marks are for your code – it will be marked for correctness and runnability.
2. Coding style: A few marks will be allocated for code style – readability, meaningful variable names, printed results, comments, etc.
3. Your understanding: The remaining marks for reflection and discussion.

We will run your code. If any parts cannot be run successfully, you will get 0 marks for that part.

Submission

The assignment must be written in **Python** in the Jupyter Notebook environment. A Jupyter Notebook template is provided. Your implementation should use the same suite of libraries that we have used during the tutorials, such as **scikit-learn**, **numpy** and **pandas**. The submission is in Canvas.

--Groups

The assignment should be completed by a group of 2-4 students. No more than 4 students are allowed.

You need to join the Canvas group: Canvas->People -> Machine Learning Project Group. The group name and number on Canvas will be your group number for assignment submission. Please make sure you and your group members are in the same Canvas group.

--Submission instructions

- When you are ready to submit your assignment, you need to submit it on behalf of the group in the corresponding submission box, **not individually**. Only one student from the group needs to submit the group assignment. When this is done, all students will be able to see the submission and later the mark (when the assignment is marked).
- You need to submit **two versions of your code: ipynb and pdf**.
- Make sure that you submit to the correct submission box! There may be penalties for incorrect submission. In summary:
 - Submit your ipynb file to "Submission: Assignment 1 ipynb";
 - Submit your pdf file to "Submission: Assignment 1 pdf";

It is important to follow the submission instructions carefully, otherwise **your mark may not be recorded correctly**.

--File names and student names

- The file names should include your **group number**, e.g. ml-project-group1.ipynb (.pdf), if you are from group 1.
- In the Jupyter Notebook, include your group name and your SIDs (as shown in the template). Do not include your name as the marking is anonymous.

A pdf of your Jupyter notebook can be generated using File>Download as>PDF or Print Preview > Save as **PDF** or File->Download as >HTML, then open the html file and save it as pdf.

Task

In this assignment, you will implement multiple classification algorithms and apply them to the **Rice dataset**. The goal is to build a complete machine learning pipeline that includes data preprocessing, classifier implementation, hyperparameter tuning, and model evaluation.

Your tasks include:

- Pre-process the dataset by replacing missing values and normalizing the values.
- Implement and evaluate the performance of multiple classification algorithms: K-Nearest Neighbour, Logistic Regression, Naïve Bayes, Decision Tree, AdaBoost, Gradient Boosting, Random Forest and Support Vector Machine using the stratified 10-fold cross-validation method. Conduct hyperparameter tuning.
- Write a short reflection and discussion report.

1. Data loading, pre-processing and printing

The dataset for this assignment is the Rice dataset, which describes two varieties of rice grains. It contains 1400 examples, described with 7 features that have been extracted from rice grain images. There are two classes – class1, corresponding to rice variety “cammeo” and class2 – to rice variety “osmancik”. The original dataset is bigger and available from the UCI Machine Learning repository: <https://archive.ics.uci.edu/dataset/545/rice+cammeo+and+osmancik>.

For this assignment, you should use the **rice-final2.csv** dataset provided on Canvas. It is a smaller and slightly modified version of the original dataset. This file includes the attribute (feature) headings, and each row corresponds to one grain example. Missing attributes in the dataset are recorded with a ‘?’.

You will need to pre-process the dataset before you can apply the classification algorithms. Three types of pre-processing are required: filling in missing values, normalization and changing the class values. After this is done, you need to print the first 10 rows of the pre-processed dataset.

1. Filling in the missing attribute values - The **missing attribute values** should be replaced with the mean value of the column using sklearn.impute.**SimpleImputer**.
2. Normalizing the data - **Normalization** of each attribute should be performed using a min-max scaler to normalize the values between [0,1] with sklearn.preprocessing.**MinMaxScaler**.
3. Changing the class values - The classes class1 and class2 should be changed to 0 and 1, respectively.
4. Print the first 10 rows of the pre-processed dataset. The feature values should be formatted to 4 decimal places using .4f; the class value is an integer. A function **print_data** has been provided in the template to help you achieve this.

Please note that we will test your code with another dataset, and your pre-processing should be written with this in mind. See the “Marking Criteria” section for more details.

2. Implement classification tasks using different classifiers

You will now apply multiple classifiers to the processed dataset, in particular: KNN, Logistic Regression, Naïve Bayes, Decision Tree, Ada Boost, Gradient Boosting, Random Forest and SVM. All classifiers should use the **sklearn** modules from the tutorials. All random states in the classifiers should be set to **random_state=0**.

Part 1: Cross-validation without parameter tuning

Implement Logistic Regression and Naïve Bayes in Part 1. Apply both classifiers to the preprocessed dataset. You need to evaluate the performance of these classifiers using 10-fold stratified cross-validation from

sklearn.model_selection.StratifiedKFold with these options:
cvKFold=StratifiedKFold(n_splits=10, shuffle=True, random_state=0)

At the end of Part 1, we should report the average cross-validation accuracy.

Part 2: Cross-validation for parameter tuning

Implement KNN, Decision Tree, Ada Boost, Gradient Boost, Random Forest and SVM in Part 2. For these classifiers, find the best parameters using grid search with 10-fold stratified cross-validation (GridSearchCV in sklearn). Find the parameters you may consider on the A1 ipynb template.

The data should be split into training and test subsets using `train_test_split` from `sklearn.model_selection` with stratification and `random_state=0`.

At the end of Part 2, we should report the best parameters for each classifier, the best cross-validation accuracy, the test set accuracy. For Random Forest, additional metrics the macro average F1 score and the weighted average F1 score are expected.

Present results for Part 1 and 2

Run the classifiers on the pre-processed dataset and print the results for each part. Format the reported results (e.g. accuracy, F1-score) to 4 decimal places using `.4f`.

3. Reflection and Discussion

In the last cell of the template, write a short reflection and discussion report that covers (but not limited to) the following aspects:

- Compare and discuss the performance of all classifiers.
- Discuss the impact of hyperparameter tuning on model performance.
- ...

Marking Criteria

The marking rubric is provided in Canvas.

We will test your program for correctness on the **rice dataset** (the given dataset) and on a second dataset (**unknown.csv**). The unknown dataset will have the same format as the rice dataset, but a different number of features and examples. Hence, do not hard-code the number of features and examples - do not set them to 1400 and 7 as in the rice dataset. For the unknown dataset, you may assume that the class variable is in the last column and that there are two classes - class1 and class2, as in the rice dataset.

To test your code before submission, we have made another dataset (**test-before.csv**) available for you. The dataset is for testing the runnability of your code, not for marking the accuracy.

In your assignment submission, your Jupyter notebook should include **only results for the rice dataset**, not the test-before dataset.

Academic honesty – very important

Please read the University policy on Academic Honesty carefully:
<https://sydney.edu.au/students/academic-integrity.html>

Plagiarism (copying from another student, website or other sources), making your work available to another student to copy, and engaging another person to complete the assignments instead of you (for payment or not) are all examples of **academic dishonesty**. Note that when there is copying between students, both students are penalized – the student who copies and the student who makes their work available for copying.

Do not confuse legitimate co-operation and cheating! You can discuss the assignment with other students, but your group must write your own code.

*You are **allowed** to use generative AI tools for this assignment.*

The University penalties are severe and include: 1) a permanent record of academic dishonesty in the university database, 2) mark deduction, ranging from 0 for the assignment to Fail for the course and 3) in severe cases, suspension from the University and cancelling of your student visa.

If there is a suspected case, the **investigation may take several months**. Your mark will not be finalized until the investigation is completed. This may create **problems enrolling in other courses next semester**

(COMP5318 is a pre-requisite for other courses) or **delaying your graduation**. Going through the investigation is also **very stressful**.

We will use similarity detection software. If you cheat, the chances that you will be caught are very high. Do not even think about engaging in plagiarism or academic dishonesty; it is not worth it. **Be smart and don't risk your reputation by engaging in plagiarism and academic dishonesty!**